



EVERLIGHT ELECTRONICS CO., LTD.

MODEL NO: 15-21VRD/TR8

Device Number : DSE-151-074 REV. 1.3

1206 Package Chip LEDs

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Features :

- Package in 8mm tape on 7" diameter reel.
- Compatible with automatic placement equipment.
- Compatible with infrared and vapor phase reflow solder process.
- Mono-color type.

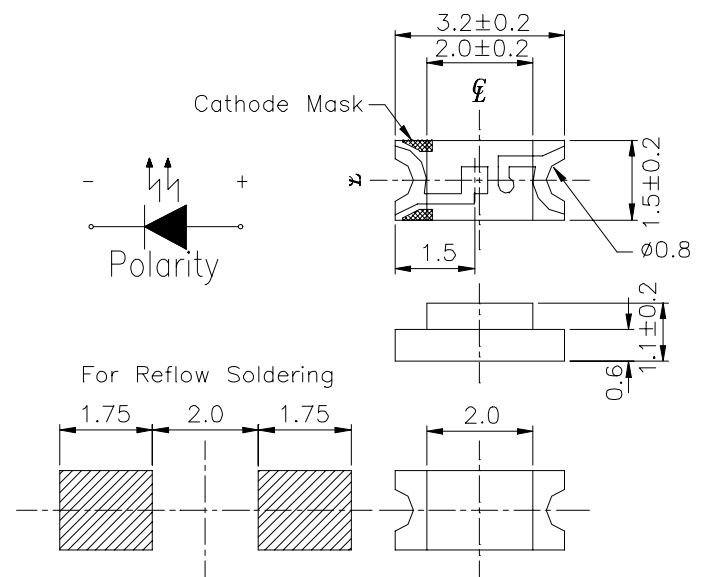
Descriptions :

- The 15-21 SMD Taping is much smaller than leaded components, thus enable smaller board size, higher packing density, reduced storage space and finally smaller equipment to be obtained.
- Besides, lightweight makes them ideal for miniature application, etc.

Applications :

- Automotive: backlighting in dashboard and switch.
- Telecommunication: indicator and backlighting in telephone and fax.
- Flat backlight for LCD, switch and symbol.
- General use.

Package Dimensions :



Notes :

Tolerances Unless Dimension $\pm$
0.1mm
Angle $\pm 0.5^\circ$
Unit = mm

PART NO	CHIP		Lens Color
	Material	Emitted Color	
15-21VRD/TR8	GaAsP/GaP	Hi-Eff Red	Red Diffused

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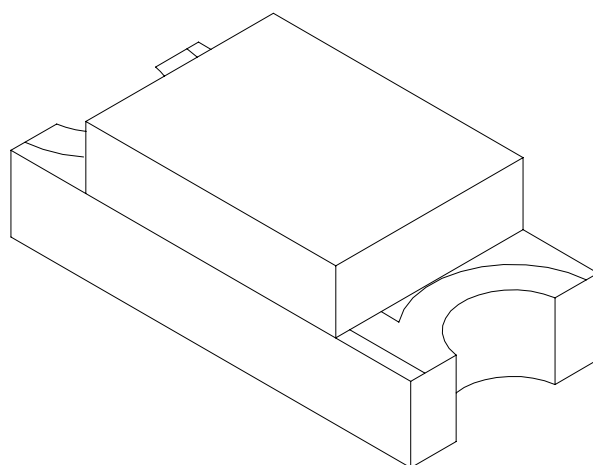


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Technical drawing of a wheel rim cross-section. The drawing shows a circular rim with a central hub. Dimensions and tolerances are indicated:

- Overall diameter: $\varnothing 180 \pm 1$
- Hub diameter: $\varnothing 50 \pm 0.5$
- Hub thickness: $11.0^{+0.5}_{-0}$
- Rim thickness: $8.4^{+1.5}_{-0}$
- Hub bore diameter: $\varnothing 12.75^{+0.25}_{-0}$
- Hub bore depth: 2.5 ± 0.5
- Rim width: 2.5 ± 0.5



Technical drawing of a progressive die. The main view shows a rectangular block with five circular holes. Dimensions include: overall width 1.65, overall height 3.55, hole diameter $\phi 1.55 \pm 0.05$, hole pitch 4.0, hole offset 1.82 ± 0.1 , hole depth 3.5 ± 0.05 , and hole spacing 2.0 ± 0.05 . A cross-section view on the right shows the internal structure with a depth of 1.65. A polarity symbol is shown on the left, and a progressive direction arrow is at the top.

TOLERANCES UNLESS DIMENSION± 0.1
ANGLE± 0.5
UNIT:mm



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■ **Absolute Maximum Ratings at Ta = 25°C**

Parameter	Symbol	Rating	Unit
Reverse Voltage	Vr	5	V
Forward Current	If	30	mA
Operating Temperature	Topr	-40 ~ +85	°C
Storage Temperature	Tstg	-40~ +90	°C
Soldering Temperature	Tsol	260 (for 5 second)	°C
Power Dissipation	Pd	100	mW
Peak Forward Current(Duty 1/10 @ 1KHZ)	If(Peak)	160	mA

■ **Electronic Optical Characteristics :**

Parameter	Symbol	Min.	Typ.	Max.	Unit	Condition
Luminous intensity	Iv	4	9	-----	mcd	If=20mA
Viewing Angle	2 θ 1/2	-----	140	-----	deg	If=20mA
Peak Wavelength	λ p	-----	640	-----	nm	If=20mA
Dominant Wavelength	λ d	-----	625	-----	nm	If=20mA
Spectrum Radiation Bandwidth	△ λ	-----	45	-----	nm	If=20mA
Forward Voltage	Vf	1.7	2.0	2.4	V	If=20mA
Reverse Current	Ir	-----	-----	10	μ A	Vr=5V



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■ Reliability Test Item And Condition

NO	Item	Test Conditions	Test Hours/Cycle	Sample Size	Ac/Re
1	Solder Heat	TEMP : 260°C ± 5 °C	5 SEC	76 Pcs	0/1
2	Temperature Cycle	H : +85°C 30min ∫ 5 min L : -55°C 30min	50 CYCLE	76 Pcs	0/1
3	Thermal Shock	H : +100°C 5min ∫ 10 sec L : -10°C 5min	50 CYCLE	76 Pcs	0/1
4	High Temperature Storage	TEMP : 100°C	1000 HRS	76 Pcs	0/1
5	Low Temperature Storage	TEMP : -55°C	1000 HRS	76 Pcs	0/1
6	DC Operating Life	If = 20 mA	1000 HRS	76 Pcs	0/1
7	High Temperature / High Humidity	85°C/85% RH	1000 HRS	76 Pcs	0/1



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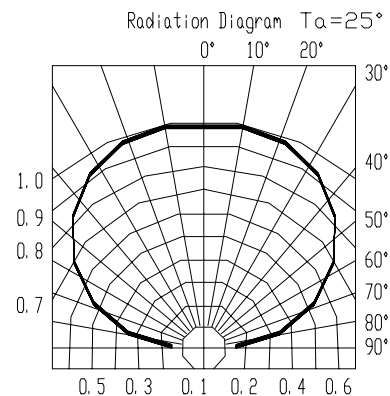
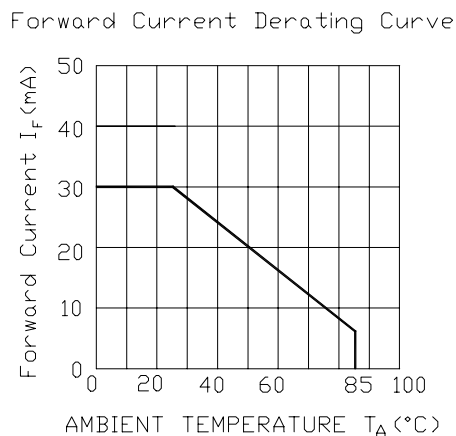
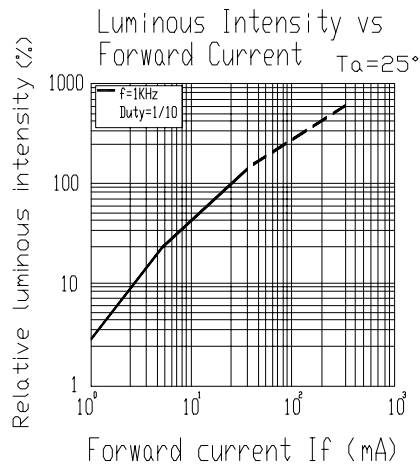
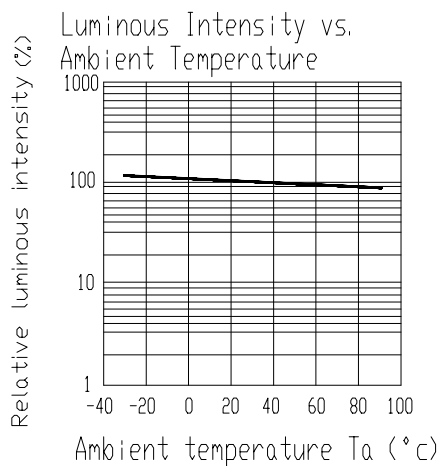
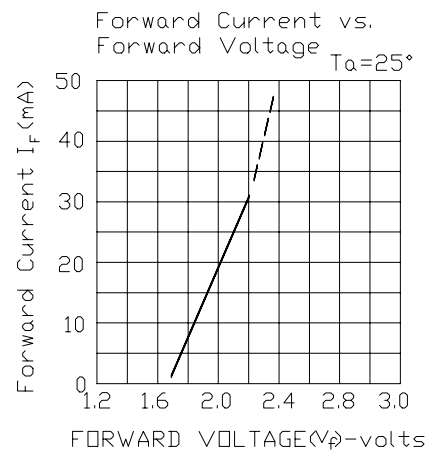
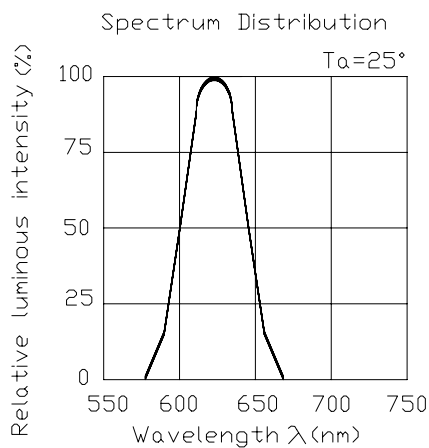
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Typical Electro-Optical Characteristic Curves





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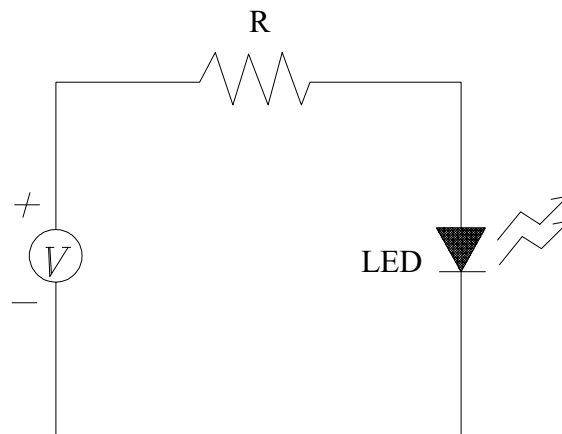
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■ Test Circuit



■ Precautions For Use

1. Over-current-proof

Customer must apply resistors for protection , otherwise slight voltage shift will cause big current change (Burn out will happen).

2. Storage time

2.1 The operation of temperature and R.H. are : $5^{\circ}\text{C}\sim 35^{\circ}\text{C}$, R.H.60%.

2.2 Once the package is opened, the products should be used within a week.

Otherwise, they should be keeping in a dampproof box with desiccants.

Considering the tape life , we suggest our customers to use our products within a year(from production date).

2.3 If opened more than one week in an atmosphere $5^{\circ}\text{C}\sim 35^{\circ}\text{C}$, R.H.60%, they should be treated at $60^{\circ}\text{C}\pm 5^{\circ}\text{C}$ for 15hrs.

2.4 When you discover that the desiccant in the package has a pink color (Normal = blue) , you should treat them in the same conditions as 2.3.



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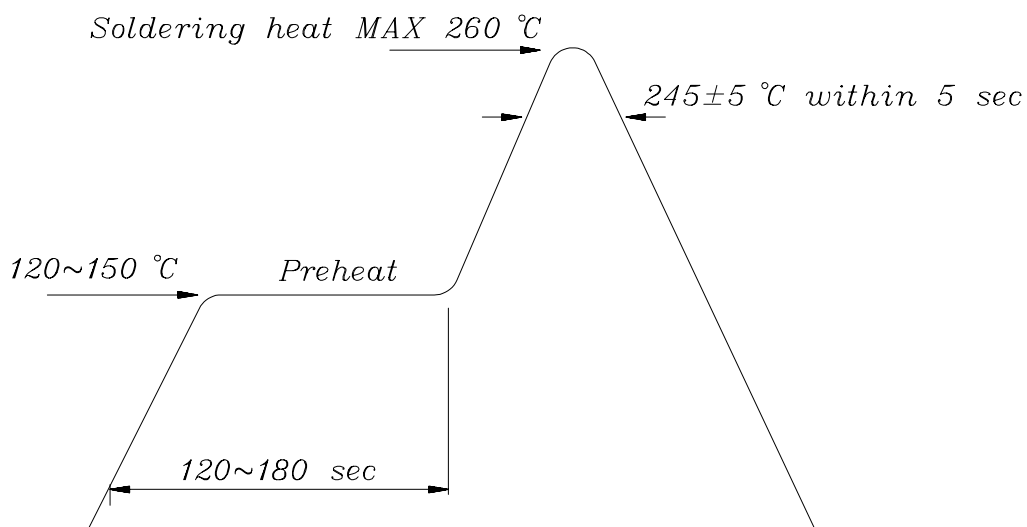
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■ Soldering heat reliability (DIP)

Please refer to the following figure :

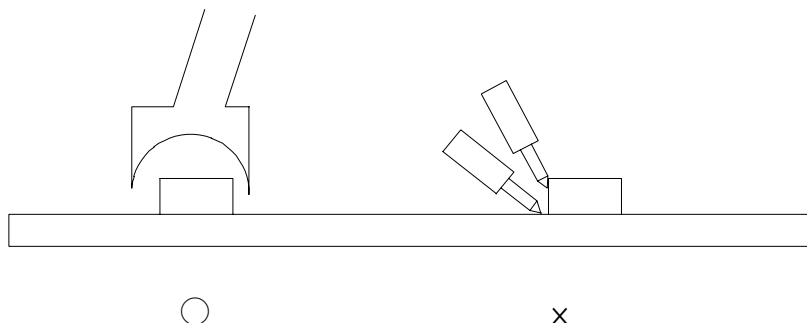


■ Soldering Iron

Basic spec is ≤ 5 sec when 260°C .If temperature is higher, time should be shorter (+10°C → -1sec). Power dissipation of iron should be smaller than 15 W , and temperature should be controllable. Surface temperature of the device should be under 230 °C .

■ Rework

1. Customer must finish rework within 5 sec under 260°C .
2. The head of iron can not touch copper foil.
3. Twin-head type is preferred.





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■ Reflow Temp / Time :

